

Representation categories of vertex algebras

ZMP Seminar

Vertex operator algebras and topological field theories from twisted QFTs in 3d and 4d

Overview

- 1 Basic definitions
- 2 The big picture of vertex algebra representation categories
- 3 How to find representations

Definition: Vertex algebra (VA)

Data:

- vector space V
- vacuum vector $\Omega \in V$
- translation operator $T : V \rightarrow V$
- field map $Y : V \otimes V \rightarrow V((z))$

Axioms:

- vacuum axiom:
 $Y(\Omega, z) = \text{id}_V$ and
 $Y(A, z)\Omega = A + zV[[z]], \forall A \in V$
- translation axiom:
 $T\Omega = 0$ and
 $[T, Y(A, z)] = \partial_z Y(A, z)$
- locality: $\forall A, B \in V, \exists n \in \mathbb{N}$
 $(z - w)^n [Y(A, z), Y(B, w)] = 0$

Consequence/Proposition

For all $A, B, C \in V$

$$Y(A, z)Y(B, w)C \in V((z))((w))$$

$$Y(B, w)Y(A, z)C \in V((w))((z))$$

$$Y(Y(A, z - w)B, w)C \in V((w))((z - w))$$

Expansion of same element
in

$$V[[z, w]][z^{-1}, w^{-1}, (z - w)^{-1}].$$

The previous slide shows vertex algebras are close to associative commutative unital $\mathbb{C}$ -algebras with a derivation. We also see that the fields $Y(A, z)$ are essentially an action of V on itself.

Definition: Vertex algebra module

Let (V, Ω, T, Y) be a vertex algebra. A V -module is a pair (M, Y_M) : M a vector space and $Y_M : V \otimes M \rightarrow M((z))$ such that

- $Y_M(\Omega, z) = \text{id}_M$
- $Y_M(TA, z) = \partial_z Y_M(A, z)$
- For all $A, B \in V$ and $C \in M$ the expansions
$$Y_M(A, z)Y_M(B, w)C \in M((z))((w))$$
$$Y_M(B, w)Y_M(A, z)C \in M((w))((z))$$
$$Y_M(Y(A, z - w)B, w)C \in M((w))((z - w))$$
can be identified in $M[[z, w]][z^{-1}, w^{-1}, (z - w)^{-1}]$.

Many additional assumptions can be added to the above definition. E.g. bounded conformal weights, finite weight spaces, semisimplicity, etc.

Let (V, Ω, T, Y) be a vertex algebra and $(M_1, Y_{M_1}), (M_2, Y_{M_2}), \dots$ be V -modules. In order to use V for conformal field theory, one needs to consider chiral correlation functions:

$$\langle Y(A, z) \phi(m_1, x_1) \phi(m_2, x_2) \cdots \rangle$$

$$\langle \phi(Y(A, z - x_1) m_1, x_1) \phi(m_2, x_2) \cdots \rangle$$

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These are essentially V -multilinear maps.

Definition: Intertwining operator

Let (V, Ω, T, Y) be a vertex algebra and $(M_1, Y_{M_1}), (M_2, Y_{M_2}), (M_3, Y_{M_3})$ be V -modules. An intertwining operator of type $\binom{M_3}{M_1, M_2}$ is a map $\mathcal{Y} : M_1 \otimes M_2 \rightarrow M_3\{x\}$ such that for all $m_i \in M_i$

- $\mathcal{Y}(m_1, z)m_2$ truncates below.

- $\mathcal{Y}(Tm_1, z) = \partial_z \mathcal{Y}(m_1, z)$.

- The expansions

$Y_{M_3}(A, z)\mathcal{Y}(m_1, x)m_2 \sim \mathcal{Y}(Y_{M_1}(A, z-x)m_1, x)m_2 \sim \mathcal{Y}(m_1, x)Y_{M_2}(A, z)m_2$
can be identified (via VA version of Jacobi identity).

Observations:

- The field map Y is an intertwining operator of type $\binom{V}{V, V}$.
- The action Y_M is an intertwining operator of type $\binom{M}{V, M}$.
- Intertwining operators are V -bilinear maps. All intertwining operators of a given type form a vector space. The field map Y and the action Y_M have a distinguished normalisation due to $Y_*(\Omega, z) = \text{id}_*$.

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Tensor products pull multilinear algebra back to linear algebra!

Definition: Fusion product aka vertex algebra tensor product

Let (V, Ω, T, Y) be a vertex algebra and $(M_1, Y_{M_1}), (M_2, Y_{M_2})$ be V -modules. A fusion product is a triple $(M_1 \boxtimes M_2, Y_{M_1 \boxtimes M_2}, \mathcal{Y}_{M_1, M_2})$, where $(M_1 \boxtimes M_2, Y_{M_1 \boxtimes M_2})$ is a V -module and $\mathcal{Y}_{M_1, M_2}$ is an intertwining operator of type $\begin{pmatrix} M_1 \boxtimes M_2 \\ M_1, M_2 \end{pmatrix}$ such that the following universal property holds: For every V -module (X, Y_X) and intertwining operator $\mathcal{Y}_X$ of type $\begin{pmatrix} X \\ M_1, M_2 \end{pmatrix}$

$$\begin{array}{ccc}
 M_1 \otimes M_2 & \xrightarrow{\mathcal{Y}_{M_1, M_2}} & M_1 \boxtimes M_2 \{z\} \\
 & \searrow \mathcal{Y}_X & \downarrow \exists! f \\
 & & X \{z\}
 \end{array}$$

In contrast to linear algebra (or ring theory) constructing $M_1 \boxtimes M_2$ and decomposing into a direct sum of indecomposable modules is *extremely* hard.

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Example: Heisenberg vertex algebra. Let F_μ be the Fock space of weight $\mu \in \mathbb{C}$. Then $\dim ({}^{F_\rho}_{F_\mu, F_\nu}) = \delta_{\rho, \mu+\nu}$ for all $\rho, \mu, \nu \in \mathbb{C}$.

$({}^{F_{\mu+\nu}}_{F_\mu, F_\nu})$ is spanned by

$$\begin{aligned} \mathcal{Y}_{F_\mu, F_\nu}(p|\mu\rangle, z)q|\nu\rangle &= z^{\mu\nu} S_\mu \prod_{m \geq 1} \exp\left(\mu \frac{a_{-m}}{m} z^m\right) Y_{F_\nu}(p|0\rangle, z) \\ &\quad \cdot \prod_{m \geq 1} \exp\left(-\mu \frac{a_m}{m} z^{-m}\right) q|\nu\rangle, \end{aligned}$$

where S_μ is the shift operator.

Well chosen categories of modules are tensor categories with respect to $\boxtimes$ with the following structures.

- For module homomorphisms $f : X \rightarrow Z$, $g : U \rightarrow W$, $f \boxtimes g$ is uniquely characterised by
 $(f \boxtimes g)\mathcal{Y}_{X,U} = \mathcal{Y}_{Z,W} \circ (f \otimes g)$
- V is the tensor identity and the unit isomorphisms are uniquely characterised by
 $\ell_M(\mathcal{Y}_{V,M}(a, z)m) = Y_M(a, z)m$ and
 $r_M(\mathcal{Y}_{M,V}(m, z)a) = e^{zT}Y_M(a, -z)m.$
- Associativity isomorphisms (hardest part!)
 $A_{M_1, M_2, M_3}(\mathcal{Y}_{M_1, M_2 \boxtimes M_3}(m_1, x_1)\mathcal{Y}_{M_2, M_3}(m_2, x_2)m_3) =$
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 All analytic details hidden.
- Braiding isomorphisms uniquely characterised by
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If the vertex algebra V is conformal (a vertex operator algebra) and the modules are chosen to be compatible with this conformal structure, then there is also a twist $\theta_M = e^{2\pi i L_0}|_M$, which satisfies the balancing equation

$$\theta_{M_1 \boxtimes M_2} = c_{M_1, M_2} \circ c_{M_2, M_1} \circ (\theta_{M_1} \boxtimes \theta_{M_2})$$

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Theorem [Huang '04]: The Verlinde Conjecture

Let (V, Ω, ω, Y) be a vertex operator algebra and $\text{Adm}(V)$ be the category of admissible V -modules. If

- 1 $\dim V_0 = 1, \dim V_{-n} = 0, \dim V_n < \infty, n \in \mathbb{N},$
- 2 V is simple as a module over itself,
- 3 $V \cong V',$ self-dual,
- 4 $\dim V/c_2(V) < \infty,$
- 5 $\text{Adm}(V)$ is semisimple,

then $\text{Adm}(V)$ is a modular tensor category. Further the action of the modular group on the category (which determines the Verlinde formula) is equal (after a renormalisation) to the action of the modular group on module characters.

Summary of what we've discussed so far

- Vertex algebras are almost commutative unital algebras with derivations.
- The conformal vector is a choice/structure: there can be 0, 1 or many.
- Vertex algebras admit modules. “Good choices” of module categories admit a tensor (aka fusion) product.
- With the exception of associators, the tensor structure morphisms follow from natural constructions and are easy to obtain.

Practical matters

- Nothing presented so far helps with actually finding modules.
- For commutative algebras, the regular module can be used to present any finitely generated module.
- This **fails** for vertex (operator) algebras. The vertex algebra as a module over itself is often assumed/required to be simple. In general almost all modules cannot be presented from sums of the vertex algebra.
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Definition/Proposition: Zhu's associative algebra, [Zhu '96]

Let (V, Ω, ω, Y) be a vertex operator algebra and consider the two binary operations

$$a \circ b = \text{Res } Y(a, z)b \frac{(1+z)^{h_a}}{z^2}, \quad a * b = \text{Res } Y(a, z)b \frac{(1+z)^{h_a}}{z}.$$

Let $O(V) = \text{span}\{a \circ b \mid \forall a, b \in V\}$. Then the following hold.

- 1 $A(V) = V/O(V)$ is a unital associative algebra under the binary operation $*$.
- 2 The class of the vacuum vector $[\Omega] = \Omega + O(V)$ is the identity element.
- 3 The class of the Virasoro vector $[\omega] = \omega + O(V)$ lies in the centre.
- 4 Let M be a V -module with ground state space $\overline{M}$. On $\overline{M}$
 $(a * b)_0 = a_0 b_0$ for all $a, b \in V$.
- 5 If $\overline{M}$ is a (left) module over $A(V)$ then it can be induced to a V -module with $\overline{M}$ as the space of ground states.

Virasoro algebra example:

- For the universal Virasoro vertex operator algebra (V_c, Ω, ω, Y) , of central charge $c \in \mathbb{C}$ we have $\mathbb{C}[X] \cong A(V_c)$, where the isomorphism is given by $X \mapsto [\omega]$.

- At minimal model central charges

$$c_{p,q} = 1 - 6 \frac{(p-q)^2}{pq}, \quad p, q \geq 2, \quad \gcd(p, q) = 1,$$

there is a singular vector $\chi \in V_{c_{p,q}}$ at degree $(p-1)(q-1)$.

Under the above isomorphism $[\chi] \in A(V_{c_{p,q}})$ corresponds to some $f(X)$ and $A(V_{c_{p,q}}/\langle \chi \rangle) \cong \mathbb{C}[X]/\langle f(X) \rangle$.

- For the Yang-Lee minimal model $p = 2, q = 5$

$$\chi = (L_{-2}^2 - \frac{3}{5}L_{-4})\Omega$$

$$Y(\chi, z) = :T(z)^2: - \frac{3}{10}\partial^2 T(z)$$

Affine example:

- Let $\mathfrak{g}$ be a complex finite-dimensional simple Lie algebra and let $V_k(\mathfrak{g})$ be the universal affine vertex operator algebra (conformal vector given by the Sugawara construction). Then $U(\mathfrak{g}) \cong A(V_k(\mathfrak{g}))$, where the isomorphism is given by $x \mapsto [x_{-1}\Omega]$.
- If $k \in \mathbb{Z}_{\geq 0}$, $(e_{-1}^\theta)^{k+1}\Omega$ is singular and generates the maximal ideal. $A(V_k(\mathfrak{g}))/\langle (e_{-1}^\theta)^{k+1}\Omega \rangle \cong U(\mathfrak{g})/\langle (e^\theta)^{k+1} \rangle$ is finite-dimensional and semisimple.
- $\mathfrak{sl}_2$ at $k = -\frac{4}{3}$. The singular vector $\chi = (h_{-3} + 3e_{-2}f_{-1} - 3e_{-1}f_{-2} + \frac{9}{2}h_{-1}e_{-1}f_{-1} + \frac{9}{8}h_{-1}^3 - \frac{9}{4}h_{-2}h_{-1})\Omega$ generates the unique non-trivial ideal.

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